The Simplicity of A_n

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In this note, we shall prove that the alternating group A_n, containing all even permutations of $\{1, \ldots, n\}$, is a simple group for $n \geq 5$. We shall first review some results about Sylow subgroups, and then present the proof that A_5 is simple. Then we shall prove by induction that A_n for $n \geq 5$ is simple.

§1 Review of Sylow theorems and some consequences

Definition 1.1. Let G be a finite group with $|G| = p^k m$ (where $p \nmid m$). A subgroup $H \leq G$ is called a **p-subgroup** if $|H| = p^i$ for some $0 \leq i \leq k$. H is called a **p-Sylow subgroup** if it's a maximal p-subgroup, i.e. if $|H| = p^k$.

Let $n_p(G)$ (or simply n_p, when the group G is clear) denote the number of p-Sylow subgroups of G, and $\mathrm{Syl}_p(G)$ denote the set of all p-Sylow subgroups of G.

Theorem 1.1 (First Sylow Theorem)

p-Sylow subgroups exist! In other words, $n_p(G) > 0$.

Theorem 1.2 (Second Sylow Theorem)

Let $K \leq G$ with $p \mid |K|$, and let H be a p-Sylow subgroup of G. Then there is a conjugate group $H' = gHg^{-1}$, such that $K \cap H'$ is a p-Sylow subgroup of K.

As a result, if K is a p-subgroup of G, then K is contained in some p-Sylow subgroup of G. Furthermore, p-Sylow subgroups are conjugate to each other!

Theorem 1.3 (Third Sylow Theorem)

Let G be a finite group with $|G| = p^k m$ (where $p \nmid m$). Then $n_p \mid m$ and $n_p \equiv 1 \pmod{p}$.

We won't go into the proofs of the Sylow theorems here. They can be found in any standard reference, for example [1]. It is, however, important to note some consequences of the Sylow theorems.

Corollary 1.4

Let H be a p-Sylow subgroup of G. Then $H \trianglelefteq G$ if and only if $n_p = 1$, i.e. it's the only p-Sylow subgroup.

Proof. If $n_p = 1$, then for any $g \in G$, $gHg^{-1} \leq G$ with

$$\left|gHg^{-1}\right| = |H|\,. \tag{1.1}$$

Hence, gHg^{-1} is also a p-Sylow subgroup of G. Since $n_p = 1$, it means they must be the same, i.e. $gHg^{-1} = H$, proving that $H \trianglelefteq G$.

Conversely, suppose $H \trianglelefteq G$. Let H' be another p-Sylow subgroup of G. Since p-Sylow subgroups are conjugate to each other (Second Sylow Theorem),

$$H' = gHg^{-1} = H \tag{1.2}$$

for some $g \in G$. The last equality follows from the normality of H. Hence, $H = H'$, proving the uniqueness of H. ■

Recall that a group G is simple if it has no nontrivial normal subgroups (except $\{e\}$ and G).

Corollary 1.5

If G is a simple group and $|G| \neq p^n$, then $n_p\,(G) > 1$ for every $p \mid G$.

Proof. If $n_p\,(G) = 1$ for some p, then the p-Sylow subgroup is normal, which means G isn't simple. ■

Corollary 1.6

If p, q are distinct prime numbers, then no group of order pq is simple.

Proof. Assume for the sake of contradiction that ther exists a simple group G with $|G| = pq$. WLOG, $p < q$. Then $n_q \mid p$. But $n_q = p$ is not possible since $n_q > 1$ along with

$$n_q \equiv 1 \pmod{q} \tag{1.3}$$

would imply

$$n_q \geq q + 1 > p. \tag{1.4}$$

Hence, $n_q = 1$, and hence the q-Sylow subgroup is normal. ■

Proposition 1.7

If G is a group of order pq and $p < q$ primes with $q \not\equiv 1 \pmod{p}$, then $G \cong \mathbb{Z}_{pq}$.

Proof. By the proof of Corollary 1.6, $n_q = 1$. $n_p \mid q$ and $q \not\equiv 1 \pmod{p}$ implies $n_p = 1$.

Therefore, $H_1 = \langle a \rangle, H_2 = \langle b \rangle$, the p-Sylow subgroup and the q-Sylow subgroup, respectively, are normal in G. Their intersection is trivial, since their orders are coprime. Therefore, a and b commute, since

$$aba^{-1}b^{-1} = \left(aba^{-1}\right)b^{-1} = a\left(ba^{-1}b^{-1}\right) \in H_1 \cap H_2 = \{e\}. \tag{1.5}$$

Now, a and b commute, their orders are coprime, so $\operatorname{ord}(ab) = pq$. Indeed,

$$\begin{aligned}(ab)^n = a^n b^n = e &\iff a^n = b^{-n} \in H_1 \cap H_2 = \{e\} &(1.6)\\ &\iff a^n = b^n = e &(1.7)\\ &\iff p \mid n \text{ and } q \mid n. &(1.8)\end{aligned}$$

So $n = pq$ is the smallest value to have $(ab)^n = e$. Now since $\operatorname{ord}(ab) = pq$, G is cyclic! ∎

Corollary 1.8

If p, q, r are distinct prime numbers, then no group of order pqr is simple.

Proof. Assume for the sake of contradiction that ther exists a simple group G with $|G| = pqr$. WLOG, $p < q < r$.

Note that, we must have $n_r > 1$, because if $n_r = 1$, then the r-Sylow subgroup would've been normal, contradicting the simplicity of G. Also, $n_r \equiv 1 \pmod{r}$ forces that $n_r > r > q > p$. Now, since

$$n_r \mid pq, \tag{1.9}$$

the only possibility is $n_r = pq$. In other words, there are pq subgroups of order r (and they intersect at the identity only, because the non-identity elements generate the whole subgroup since it's a cyclic group), which gives

$$(\text{number of elements of order } r) = n_r(r-1) = pq(r-1) = |G| - pq. \tag{1.10}$$

Similarly, we must have $n_q > 1$, because if $n_q = 1$, then the q-Sylow subgroup would've been normal, contradicting the simplicity of G. Also, $n_q \equiv 1 \pmod{q}$ forces that $n_q > q > p$. Therefore,

$$(\text{number of elements of order } q) = n_q(q-1) > q^2 - q. \tag{1.11}$$

Therefore,

$$\begin{aligned} n_p\,(p-1) &= (\text{number of elements of order } p) && (1.12)\\ &< pq - \left(q^2 - q\right) && (1.13)\\ &= q\,(p-q+1) \leq 0. && (1.14) \end{aligned}$$

This can't be the case since $n_p \geq 1$. Therefore, there is no simple group of order pqr. ■

Corollary 1.9

There is no simple group of order 12.

Proof. Assume the contrary. Note that $12 = 2^2 \times 3$. Since $n_3 \mid 4$ and $n_3 > 1$ (since we assumed the group is simple) and $n_3 \equiv 1 \pmod 3$, we must have that $n_3 = 4$. This gives

$$(\text{number of elements of order } 3) = n_3(3-1) = 4 \cdot 2 = 8. \tag{1.15}$$

The 2-Sylow subgroups can't intersect any of the 3-Sylow subgroups. The identity element together with the other 3 elements form one 2-Sylow subgroup. Therefore, we don't have any room for any other 2-Sylow subgroups, implying $n_2 = 1$. ■

Corollary 1.10

If p, q are distinct primes, there is no simple group of order p^2q.

Proof. Assume for the sake of contradiction that ther exists a simple group G with $|G| = p^2q$. Note that

$$p + 1 \leq n_p \leq q. \tag{1.16}$$

The first bound follows from the fact that $n_p \neq 1$, and the second bound is given by $n_p \mid q$. Now, note that $n_q \mid p^2$ and $n_q > q > p$, so we must have $n_q = p^2$. This gives us

$$p^2 \equiv 1 \pmod q \implies q \mid (p-1)\,(p+1)\,. \tag{1.17}$$

Since $q > p$, the scenario $q \mid p-1$ is impossible, so we must have $q \mid p+1$. But that way $q \leq p+1$. But we already know $q \geq p+1$. Therefore, $q = p+1$. This can only happen if $p = 2$, $q = 3$, i.e. $G = 12$, which has already been done! ■

Remark 1.1. It's important to note that in all the cases that we worked with, e.g. $|G| = pq, pqr, p^2q$ etc, the key contradiction was by showing that $n_p = 1$ for some prime p. Therefore, groups of these order are not only simple, they have a normal p-Sylow subgroup!

Proposition 1.11

Let H be a p-Sylow subgroup of G, and $N_G(H)$ is the normalizer of H in G. Then

$$n_p = [G : N_G(H)]. \tag{1.18}$$

Proof. Consider the action of G on $\mathrm{Syl}_p(G)$ by conjugation. Then

$$N_G(H) = \{g \in G \mid gHg^{-1} = H\} \tag{1.19}$$

is nothing but $\mathrm{Stab}(H)$. Therefore, by Orbit-Stabilizer theorem (Lemma 5.12 from [1]),

$$n_p = \left|\mathrm{Syl}_p(G)\right| = |\mathrm{Orb}(H)| = \frac{|G|}{|\mathrm{Stab}(H)|} = [G : N_G(H)]. \tag{1.20}$$

■

Proposition 1.12

If $N \trianglelefteq G$, and $H \trianglelefteq N$ is a p-Sylow subgroup of N, then $H \trianglelefteq G$.

Proof. Since $H \trianglelefteq N$, $n_p(N) = 1$. Then for any $g \in G$,

$$gHg^{-1} \subseteq gNg^{-1} = N. \tag{1.21}$$

Hence, gHg^{-1} is a p-Sylow subgroup of N, since $\left|gHg^{-1}\right| = |H|$. But since $n_p(N) = 1$, we must have that $gHg^{-1} = H$, for any $g \in G$. As a result, $H \trianglelefteq G$. ■

Corollary 1.13

A group of order 30 contains a normal 5-Sylow subgroup.

Proof. By the proof of Corollary 1.8, $n_5 > 1$ and $n_3 > 1$ both can't happen. If $n_5 = 1$, then we are done! So suppose $n_3 = 1$. Let H_3 be the 3-Sylow subgroup and H_5 be a 5-Sylow subgroup. Consider $H = H_3H_5 \leq G$. Then

$$|H| = \frac{|H_3|\,|H_5|}{|H_3 \cap H_5|} = 15, \tag{1.22}$$

and from Proposition 1.7, we conclude that H is cyclic. Since $[G : H] = 2$, H is normal in G. Furthermore, since H is cyclic, it has a unique 5-Sylow subgroup, which is H_5. Therefore, by Proposition 1.12, $H_5 \trianglelefteq G$, i.e. $n_5(G) = 1$. ■

Corollary 1.14

A group of order 20 contains a normal 5-Sylow subgroup.

Proof. Since $n_5 \mid 4$ and $n_5 \equiv 1 \pmod 5$, we must have $n_5 = 1$. ■

§2 Proof that A_5 is simple

Theorem 2.1

A_5 is simple.

Proof. Assume for the sake of contradiction that $G := A_5$ is not simple, i.e. it has a normal subgroup H such that $\{e\} \lneq H \lneq G$. Then since $|H|$ divides $|G| = 60$, the possible values for $|H|$ are

$$|H| \in \{2, 3, 4, 5, 6, 10, 12, 15, 20, 30\}. \tag{2.1}$$

Claim 2.1: $n_5 = 6$.

Proof. $|G| = 60 = 5 \cdot 12$. So we have $n_5 \mid 12$. This together with $n_5 \equiv 1 \pmod 5$ gives us $n_5 \in \{1, 6\}$. But the 5-Sylow subgroups

$$\langle (1\ 2\ 3\ 4\ 5) \rangle \text{ and } \langle (1\ 3\ 2\ 4\ 5) \rangle \tag{2.2}$$

are distinct. Hence, $n_5 \geq 2$, so that $n_5 = 6$. □

Claim 2.2: 5 doesn't divide $|H|$.

Proof. Assume the contrary. Then H contains a 5-Sylow subgroup. Since H is normal in G, and the 5-Sylow subgroups are conjugate to each other, H must contain all 5-Sylow subgroups. Then the 5-Sylow subgroups intersect only at the identity, and hence

$$|H| \geq n_5 (5 - 1) + 1 = 25; \tag{2.3}$$

because there are $n_5 (5 - 1)$ elements of order 5, and then $+1$ accounting for the identity. Then (2.1) gives us that $|H| = 30$. But then Corollary 1.13 implies that there exists a unique 5-Sylow subgroup of H, which can't be true since H contains all six 5-Sylow subgrops. □

Therefore,

$$|H| \in \{2, 3, 4, 6, 12\}. \tag{2.4}$$

If $|H| = 6$ or 12, it has a normal 2-Sylow subgroup, or a normal 3-Sylow subgroup (Corollary 1.6 and Corollary 1.9). These normal Sylow subgroups are of size $2, 3$, or 4. Since normal Sylow subgroups of H are also normal subgroups of G (by Proposition 1.12), it suffices to consider

$$|H| \in \{2, 3, 4\}. \tag{2.5}$$

For $|H| = 2, 3, 4$, consider $\overline{G} = G/H$, which has order $30, 20, 15$, respectively. Each of them has a normal 5-Sylow subgroup (Corollary 1.13, Corollary 1.14, Proposition 1.7), let's call it $\overline{H_5}$. Consider the quotient homomorphism

$$\pi : G \to \overline{G} = G/H. \tag{2.6}$$

The preimage of a normal subgroup under a group homomorphism is normal, so $\pi^{-1}\left(\overline{H_5}\right)$ is a normal subgroup of G. But since $\left|\overline{H_5}\right| = 5$, we have $5 \mid \left|\pi^{-1}\left(\overline{H_5}\right)\right|$. But we have already shown that no normal subgroup of G has order divisible by 5 (Claim 2.2). Thus we arrive at a contradiction! ■

§3 Alternate proof using the cycle structure

Lemma 3.1

For $n \geq 5$, any two 3-cycles are conjugate in A_n.

Proof. It suffices to show that any 3-cycle is conjugate to $(1\ 2\ 3)$. Let $\sigma \in A_n$ be a 3-cycle. Then there exists some $\tau \in S_n$ such that

$$\tau\sigma\tau^{-1} = (1\ 2\ 3)\,. \tag{3.1}$$

If $\tau \in A_n$, we are done! Otherwise, consider $\rho := (4\ 5)\,\tau \in A_n$.

$$\rho\sigma\rho^{-1} = (4\ 5)\,\tau\sigma\tau^{-1}\,(4\ 5) = (4\ 5)\,(1\ 2\ 3)\,(4\ 5) = (1\ 2\ 3)\,. \tag{3.2}$$

■

Lemma 3.2

For $n \geq 3$, A_n is generated by the 3-cycles.

Proof. Let T be the set of all 3-cycles. Then clearly $\langle T \rangle \subseteq A_n$. Now take $\sigma \in A_n$. Then it's the product of an even number of transpositions. So it suffices to show that $(a\ b)\,(c\ d)$ is generated from transpositions. We have the following cases:

1. If a, b, c, d are pairwise distinct: then

$$(a\ b)\,(c\ d) = (a\ c\ b)\,(a\ c\ d)\,. \tag{3.3}$$

2. If they are not pairwise distinct, we can't have $a = b$ or $c = d$. Suppose $a = c$. Then

$$(a\ b)\,(a\ d) = (a\ d\ b)\,. \tag{3.4}$$

The cases $b = d$, $a = d$, $b = c$ are all the same!

Therefore, $\sigma \in \langle T \rangle$, i.e. $\langle T \rangle = A_n$. ■

Theorem 3.3

A_5 is simple.

Proof. Assume for the sake of contradiction that $G := A_5$ is not simple, i.e. it has a normal subgroup H such that $\{e\} \lneq H \lneq G$. The cycle structure of the elements of A_5 are one of the following:

$$(a\ b\ c)\,, (a\ b)\,(c\ d)\,, (a\ b\ c\ d\ e)\,. \tag{3.5}$$

If H contains a 3-cycle, then it would contain all the 3-cycles since they are all conjugate to each other (Lemma 3.1). Then H would be the whole A_n, by Lemma 3.2. Therefore, H can't contain any 3-cycles.

If H contains a 5-cycle $(a\ b\ c\ d\ e)$, then it also contains

$$\begin{aligned} &\Big[(a\ b\ c)\,(a\ b\ c\ d\ e)\,(a\ b\ c)^{-1}\Big]\,(a\ b\ c\ d\ e)^{-1} \\ =\ &(a\ b\ c)\,(a\ b\ c\ d\ e)\,(a\ c\ b)\,(a\ e\ d\ c\ b) = (a\ b\ d) \end{aligned} \tag{3.6}$$

If H contains $(a\ b)\,(c\ d)$, then it also contains

$$\begin{aligned} &\Big[(a\ b\ e)\,(a\ b)\,(c\ d)\,(a\ b\ e)^{-1}\Big]\,(a\ b)\,(c\ d) \\ =\ &(a\ b\ e)\,(a\ b)\,(c\ d)\,(a\ e\ b)\,(a\ b)\,(c\ d) = (a\ e\ b) \end{aligned} \tag{3.7}$$

Therefore, H always contains a 3-cycle. Hence, $H = A_5$. In other words, no nontrivial normal subgroups of A_5 exists. ■

§4 Alternate proof by counting cycles

For a prime p, an element $\sigma \in S_n$ has order p if and only if in the disjoint cycle decomposition of σ, the disjoint cycles are all of length p. So let's count how many p-cycles are there in S_n.

Lemma 4.1

The number of p-cycles in S_n is

$$\binom{n}{p}\,(p-1)\,! \tag{4.1}$$

Proof. We can choose the p numbers belonging in the p-cycle in $\binom{n}{p}$-many ways. But each cyclic permutation of $(a_1, \ldots, a_p)$ gives rise to a new p-cycle. And there are $(p-1)\,!$-many cyclic permutations. Hence, The number of p-cycles in S_n is $\binom{n}{p}\,(p-1)\,!$. ■

Therefore,

- the number of 5-cycles in A_5, i.e. the number of elements of order 5 is $\binom{5}{5}4! = 24$;
- the number of 3-cycles in A_5, i.e. the number of elements of order 3 is $\binom{5}{3}2! = 20$;
- the elements of order 2 in A_5 are $(a\ b)\,(c\ d)$, so the number of elements of order 2 is

$$\frac{1}{2}\binom{5}{2}\binom{3}{2} = 15, \tag{4.2}$$

since we can pick a, b in $\binom{5}{2}$-many ways, and c, d in $\binom{3}{2}$-many ways, and $(a\ b)\,(c\ d)$ is the same as $(c\ d)\,(a\ b)$, so we need to divide by 2.

Lemma 4.2

Let G be a finite group and $N \trianglelefteq G$, and $g \in G$. If

$$\gcd\left(\mathrm{ord}(g), |G/N|\right) = 1, \tag{4.3}$$

then $g \in N$.

Proof.

$$(gN)^{\mathrm{ord}(g)} = g^{\mathrm{ord}(g)}N = N \quad \text{and} \quad (gN)^{|G/N|} = N. \tag{4.4}$$

Therefore,

$$\mathrm{ord}\,(gN) \mid \mathrm{ord}(g) \text{ and } \mathrm{ord}\,(gN) \mid |G/N|\,. \tag{4.5}$$

As a result,

$$\mathrm{ord}\,(gN) \mid \gcd\left(\mathrm{ord}(g), |G/N|\right) = 1. \tag{4.6}$$

Hence, $gN = N$, so $g \in N$. ■

Theorem 4.3

A_5 is simple.

Proof. Assume for the sake of contradiction that $G := A_5$ is not simple, i.e. it has a normal subgroup H such that $\{e\} \lneq H \lneq G$. Then since $|H|$ divides $|G| = 60$, the possible values for $|H|$ are

$$|H| \in \{2, 3, 4, 5, 6, 10, 12, 15, 20, 30\}\,. \tag{4.7}$$

We divide the proof into several steps:

- $\boxed{|H| \neq 30}$: If $|H| = 30$, then $|G/H| = 2$, so it's coprime with 3 and 5. Hence, all the elements of order 3 and 5 are in H, by Lemma 4.2. But there are $24 + 20 = 44$ of them, whereas $|H| = 30$.
- $\boxed{|H| \neq 4}$: If $|H| = 4$, then $|G/H| = 15$, so it's coprime with 2. Hence, all the elements of order 2 are in H, by Lemma 4.2. But there are 15 of them, whereas $|H| = 4$.

- $\boxed{3 \nmid |H|}$: If $3 \mid |H|$, $3 \nmid |G/H|$, so by Lemma 4.2, H contains all elements of order 3. There are 20 of them, but $|H| \leq 15$.
- $\boxed{5 \nmid |H|}$: If $5 \mid |H|$, $5 \nmid |G/H|$, so by Lemma 4.2, H contains all elements of order 5. There are 24 of them, but $|H| \leq 20$.

Now, the only case left is $|H| = 2$. Then $|G/H| = 30$, so G/H contains a nontrivial normal subgroup (Corollary 1.8), let's call it $\overline{N}$. Consider the quotient homomorphism

$$\pi : G \to G/H. \tag{4.8}$$

The preimage of a normal subgroup under a group homomorphism is normal, so $\pi^{-1}\left(\overline{N}\right) = NH$ is a nontrivial normal subgroup of G. But $|NH| \neq 2 = |H|$, and we have already shown that the only possible order of normal subgroup of G is 2. Thus, we arrive at a contradiction! ■

§5 Alternate proof using conjugacy classes

This is similar to the previous proof in spirit. We count the number of elements in each conjugacy class of A_5. We have already shown that any two 3-cycles are conjugate to each other (Lemma 3.1).

Lemma 5.1

For $n \geq 4$, any two permutations of cycle-type $(2, 2)$ are conjugate in A_n.

Proof. It suffices to show that any $\sigma = (a\ b)\,(c\ d)$ is conjugate to $(1\ 2)\,(3\ 4)$. Since σ is conjugate to $(1\ 2)\,(3\ 4)$ in S_n, there exists some τ such that

$$\tau\sigma\tau^{-1} = (1\ 2)\,(3\ 4)\,. \tag{5.1}$$

If $\tau \in A_n$, we are done! Otherwise, consider $\rho = (1\ 2)\,\tau$. Then

$$\rho\sigma\rho^{-1} = (1\ 2)\,\tau\sigma\tau^{-1}\,(1\ 2) = (1\ 2)\,(1\ 2)\,(3\ 4)\,(1\ 2) = (1\ 2)\,(3\ 4)\,, \tag{5.2}$$

since $(1\ 2)$ and $(3\ 4)$ commute. ■

Any two 5-cycles are conjugate in S_5. But that need not be the case in A_5.

Lemma 5.2

$\sigma_1 = (1\ 2\ 3\ 4\ 5)$ and $\sigma_2 = (1\ 3\ 2\ 4\ 5)$ are not conjugate in A_5.

Proof. σ_1 and σ_2 are conjugate in S_5, we we can find $\tau \in S_5$ such that

$$\sigma_2 = \tau\sigma_1\tau^{-1}. \tag{5.3}$$

In fact, $\tau = (2\ 3)$ is one such permutation, but it's not in A_5. Suppose $\rho \in S_5$ is another such element.

$$\sigma_2 = \tau\sigma_1\tau^{-1} = \rho\sigma_1\rho^{-1} \implies \left(\tau^{-1}\rho\right)\sigma_1 = \sigma_1\left(\tau^{-1}\rho\right). \tag{5.4}$$

Therefore, $\tau^{-1}\rho \in C_{S_5}(\sigma_1)$.

Claim 5.1: $C_{S_5}(\sigma_1) = \langle\sigma_1\rangle \leq A_5$.

Proof. Let G act on itself by conjugation. Then $\mathrm{Orb}(g)$ is the conjugacy class $\mathrm{Conj}_G(g)$ of g, and

$$\mathrm{Stab}(g) = \left\{x \in G \mid xgx^{-1} = g\right\} = \{x \in G \mid xg = gx\} = C_G(g). \tag{5.5}$$

By Orbit-Stabilizer theorem,

$$|C_G(g)| = |\mathrm{Stab}(g)| = \frac{|G|}{|\mathrm{Orb}(g)|} = \frac{|G|}{|\mathrm{Conj}_G(g)|}. \tag{5.6}$$

Taking $g = \sigma_1$ in $G = S_5$, we get that the conjugacy class of σ_1 consists of all 5-cycles, and hence $|\mathrm{Conj}_G(g)| = 4! = 24$. Therefore,

$$|C_{S_5}(\sigma_1)| = \frac{120}{24} = 5. \tag{5.7}$$

Since $\langle\sigma_1\rangle$ is a cyclic group, its elements commute with σ_1. Hence, $C_{S_5}(\sigma_1) = \langle\sigma_1\rangle$. □

Hence, $\tau^{-1}\rho \in A_5$. Therefore, $\rho \in \tau A_5$, so ρ can never be an even permutation. Therefore, σ_1 and σ_2 are not conjugate in A_5. ■

Lemma 5.3

$\left|\mathrm{Conj}_{A_5}(\sigma_1)\right| = \left|\mathrm{Conj}_{A_5}(\sigma_2)\right| = 12$.

Proof. By (5.6),

$$\left|\mathrm{Conj}_{A_5}(\sigma_i)\right| = \frac{|A_5|}{|C_{A_5}(\sigma_i)|}. \tag{5.8}$$

All the elements of S_5 that commute with σ_i are $C_{S_5}(\sigma_i) = \langle\sigma_i\rangle \leq A_5$ (Claim 5.1). Therefore, all the elements of A_5 that commute with σ_i are the same ones! Hence,

$$|C_{A_5}(\sigma_i)| = 5. \tag{5.9}$$

Therefore,

$$\left|\mathrm{Conj}_{A_5}(\sigma_i)\right| = \frac{60}{5} = 12, \tag{5.10}$$

for $i = 1, 2$. ■

Therefore, the conjugacy classes of A_5 are:

1. elements of cycle type $(1, 1, 1, 1, 1)$, i.e. the identity e only;

2. elements of cycle type $(3,1,1)$, i.e. the 3-cycles, and there are 20 of them (Lemma 4.1);
3. elements of cycle type $(2,2,1)$, i.e. the product of two disjoint transpositions, and there are 15 of them, by (4.2);
4. elements of cycle type (5) that are conjugate to $(1\ 2\ 3\ 4\ 5)$, and there are 12 of them, by Lemma 5.3;
5. elements of cycle type (5) that are conjugate to $(1\ 3\ 2\ 4\ 5)$, and there are 12 of them, by Lemma 5.3.

These are all the conjugacy classes since $1+20+15+12+12=60$.

Theorem 5.4

A_5 is simple.

Proof. Assume for the sake of contradiction that $G := A_5$ is not simple, i.e. it has a normal subgroup H such that $\{e\} \lneq H \lneq G$. Then since $|H|$ divides $|G| = 60$, the possible values for $|H|$ are

$$|H| \in \{2,3,4,5,6,10,12,15,20,30\}. \tag{5.11}$$

Normal subgroups are made of conjugacy classes (i.e. if it contains one element of the conjugacy class, it contains all the elements of the conjugacy class). So we summarize the conjugacy classes of A_5:

Representative of conjugacy class	e	$(1\ 2\ 3\ 4\ 5)$	$(1\ 3\ 2\ 4\ 5)$	$(1\ 2)(3\ 4)$	$(1\ 2\ 3)$
Order of conjugacy class	1	12	12	15	20

H contains e. Since H is nontrivial, it contains at least one of the conjugacy classes, so $|H| > 12$. So $|H| = \{15, 20, 30\}$. But adding the order of the conjugacy classes, we can't get to $15, 20$, or 30. Hence, no nontrivial normal subgroups of A_5 exist! ∎

§6 Generalizing to A_n for $n \geq 6$

In this section, we shall prove that A_n is simple for $n \geq 5$. Notice that $|A_3| = 3$, so it's a simple abelian group. Also, $|A_4| = 12$, so it can't be simple by Corollary 1.9.

Lemma 6.1

When $n \geq 5$, each nontrivial $\sigma \in A_n$ has a conjugate $\sigma' \neq \sigma$ such that $\sigma(i) = \sigma'(i)$ for some i.

For example, if $\sigma = (1\ 2\ 3\ 4\ 5)$ in A_5, then $\sigma' = (3\ 4\ 5)\sigma(3\ 4\ 5)^{-1} = (1\ 2\ 4\ 5\ 3)$ has the same value at $i = 1$ as σ does.

Proof. Let r be the longest length of a disjoint cycle in σ. By replacing σ with a conjugate permutation (which is also in A_n and has the effect of relabeling the numbers from 1 to n when σ permutes them), we can assume the disjoint r-cycle in σ is $(1\ 2\ \cdots\ r)$, and then we can write

$$\sigma = (1\ 2\ \cdots\ r)\pi, \tag{6.1}$$

where $(1\ 2\ \cdots\ r)$ and π are disjoint.

If $r \geq 3$, let $\tau = (3\ 4\ 5)$ and $\sigma' = \tau\sigma\tau^{-1}$. Then

$$\sigma(1) = 2,\ \sigma'(1) = 2, \quad \sigma(2) = 3,\ \sigma'(2) = 4. \tag{6.2}$$

Thus $\sigma' \neq \sigma$ and both take the same value at 1. If $r = 2$, then σ is a product of disjoint transpositions. If there are at least 3 disjoint transpositions involved, then $n \geq 6$, and we can write

$$\sigma = (1\ 2)(3\ 4)(5\ 6)\cdots \tag{6.3}$$

after relabeling. Let $\tau = (1\ 2)(3\ 5)$ and $\sigma' = \tau\sigma\tau^{-1}$. Then

$$\sigma(1) = 2,\ \sigma'(1) = 2, \quad \sigma(3) = 4,\ \sigma'(3) = 6. \tag{6.4}$$

Again, we see $\sigma' \neq \sigma$ and σ and σ' have the same value at 1. If $r = 2$ and σ is a product of 2 disjoint transpositions, write $\sigma = (1\ 2)(3\ 4)$ after relabeling. Let $\tau = (1\ 3\ 2)$ and

$$\sigma' = \tau\sigma\tau^{-1} = (1\ 3)(2\ 4). \tag{6.5}$$

Then $\sigma' \neq \sigma$ and they both fix 5. ∎

Theorem 6.2

For $n \geq 5$, A_n is simple.

Proof. We proceed by induction on n. The base case $n = 5$ has already been done, so we may assume $n \geq 6$.

Consider the action of A_n on $\{1, 2, \ldots, n\}$, and let $H_i \leq A_n$ be the subgroup that fixes i. Then

$$H_i \cong A_{n-1}, \tag{6.6}$$

so by inductive hypothesis, each H_i is simple. Furthermore, note that each H_i contains a 3-cycle (just take the 3-cycle $(a_1\ a_2\ a_3)$, where none of the numbers are i).

Let $N \trianglelefteq A_n$ be a nontrivial normal subgroup. We want to show that $N = A_n$. Take $\sigma \in N$ with $\sigma \neq e$. By Lemma 6.1, there is a conjugate σ' of σ such that $\sigma' \neq \sigma$ and $\sigma(i) = \sigma'(i)$ for some i.

Since N is normal in A_n, $\sigma' \in N$. Then $\sigma^{-1}\sigma'$ is a non-identity element of N that fixes i, so $N \cap H_i$ is a nontrivial subgroup of H_i. It is also a normal subgroup of H_i since $N \trianglelefteq A_n$. Since

H_i is simple,

$$N \cap H_i = H_i, \tag{6.7}$$

i.e. $H_i \subseteq N$. Since H_i contains a 3-cycle, N contains a 3-cycle. By the normality of N, it contains all 3-cycles. Since 3-cycles generate A_n, $N = A_n$, and we are done! ■

From $H_i \subseteq N$, there's another way of finishing without looking at the 3-cycles. Note that different H_i's are conjugate subgroups of A_n, since

$$\sigma H_i \sigma^{-1} = H_{\sigma(i)}. \tag{6.8}$$

Therefore, N contains every $H_{\sigma(i)}$ for all $\sigma \in A_n$. Since $\sigma(i)$ can be an arbitrary element of A_n as σ varies in A_n, N contains every H_i.

Every permutation of cycle-type $(2, 2)$ is in some H_i since $n \geq 5$, so N contains all permutations of cycle-type $(2, 2)$. Every permutation in A_n is a product of permutations of type $(2, 2)$, so $N \supseteq A_n$. Therefore $N = A_n$.

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